LEARN CODING

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github.com/ale66/learn-coding

WELCOME

- Introduction 👋
- Materials ⬇️

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PLAN

1. What is Coding? Why should I learn it?
2. A tour of computers
3. A tour of [online] learning resources
4. A peek at topics

WHAT IS CODING?

WHAT IS CODING?

A creative activity where computers are instructed directly to perform useful, step-by-step operations.

YES, BUT HOW?

In music we go from *imagined* sounds to music score to execution (press piano keys, blow air in trumpet, harp on strings) to actual, perceived sounds.

Humans communicate on three levels: natural language, the music score and the execution.

METHAPHORICALLY

- the music score represents the code.
- execution on a specific instrument represents executable code, e.g., file chrome.exe on your computer
- the hearing *experience* represents the changes that take place on your data.

Coding in some ways is like music composition

UN-METAPHORICALLY...

- an informal language will describe algorithms on paper, on whiteboard etc.
- a formal language, Python or SQL or Markdown, will describe code.
- special interpretation/compilation software will take code and execute it.
- we need to supply *data* and store results.

ALGORITHMS, BY EXAMPLE

It's nice and clear, and 25 degrees in Naples today. No, it's 77 in Naples (FL)

the Farenheit temperature in Naples can be obtained by

1. taking the current temperature in Celsius degrees
2. rescale it by $\frac{9}{5}$
3. re-center it by adding 32

CODE

```
1  my_celsius = int(input('Please enter the current temperature in Naples:'))
2
3  my_farenheit = (my_celsius * 9/5) + 32
4
5  print(f'Today we have {my_farenheit} Farenheit degrees in Naples')
```

Reading this syntax requires training.

IMPLEMENTATION: COMPILATION

```
1  >python -m py_compile my_converter.py -o converter.exe
2  >converter.exe
3  >Please enter the temperature in Naples now:
```

file converter.exe is for computers, not humans.

IMPLEMENTATION: INTERPRETATION

```
1  >python my_converter.py
2  >Please enter the current temperature in Milan:
```

COMPUTERS

COMPUTERS

Electronics, networks

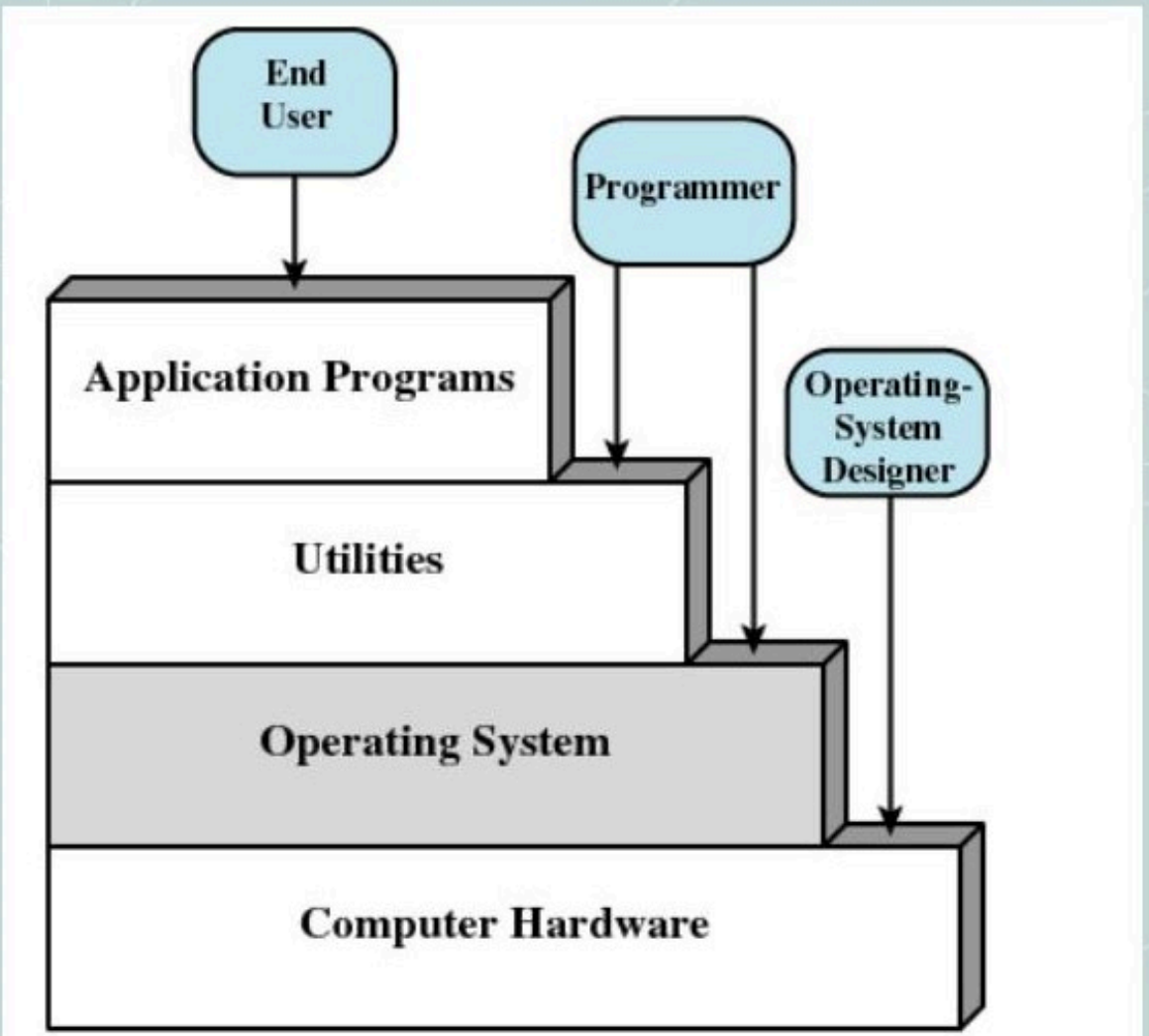
Operating system

Software

OPERATING SYSTEMS

- computers/smartphones come with a pre-loaded set of executable files that create the *operating system environment*
- create an abstract view of the computer: specific hardware details are now trasparent
- special abstraction: **the file system**
- all of them offer basic functionalities for coding
- let's get started with the **file system** and the **command line interface**

Layers of computer system



AND WHAT'S A FILE?

- computer memory is best seen as a long ribbon where, at different times, we write sequences of bytes, called *files*
- a file is i) a unique name, ii) a memory address inside the compuhardware iii) a sequence of bits, the actual content, iv) ownership information (for later) and v) a format that guides the interpretation of the bits: are they color pixels? Characters? Numbers?

THE FILE SYSTEM

- the Os shows files are organised in a hyerachical structure of *manila folders*



THE FILE SYSTEM

- a special file, called folder, contains the names and physical addresses of the files within
- each folder contains two special (and secret) files: `.` and `..`
- file `.` contains a reference to the actual positioning of the folder on the memory device
- file `..` contains a reference to the *containing* folder

THE CLI: COMMAND-LINE INTERFACE

- iOs: `Terminal`
- Win: `Cmd` or `Powershell` or `Windows terminal`
- Linux/Android: `sh` or `bash`

Structure:

```
>pwd
>dir
>notepad my_converter.py
>cd ..
>dir
>pwd
>notepad my_converter.py
```

PATHS

File must have unique names

No repeated names in the same folder

Ok to repeated names in different folders, how?

relative path: `.\my_converter.py`

the absolute path: `C:\Users\ale\_git\learn-coding\00-get_started+basic_os\converter.py`

another abs. path: `C:\Users\ale\Downloads\converter.py`

MORE RELATIVE PATHS:

Mac:

```
1  >pwd
2  /home/ale/_git/learn-coding/02-get_started+syntax
3  >python ../00-get_started+basic_os/example-temperature-converter/my_converter.py
```

Win:

```
1  >cd
2  C:\Users\aless\Downloads\_git\learn-coding\02-get_started+syntax
3  >python ..\00-get_started+basic_os\example-temperature-converter\my_converter.py
```

TRY IT OUT

Win:

Start `cmd` or `Windows terminal`

```
1  >cd ..
2  >dir
3  >notepad my_converter.py
```

Mac:

Start the Terminal app from the tray

```
1  >cd ..
2  >pwd
3  >ls -lrt
4  >open /System/Applications/TextEdit.app
5  >open -e my_converter.py
```